



2002-01-03

PRODUKTINFORMATION

Vi reserverar oss mot fel samt förbehåller oss rätten till ändringar utan föregående meddelande

ELFA artikelnr

73-217-30 ST3232CN RS 232 krets

73-217-48 ST3232CD RS232 krets



ST3232

3 TO 5.5V, LOW POWER, UP TO 400KBPS, RS-232 DRIVERS AND RECEIVERS

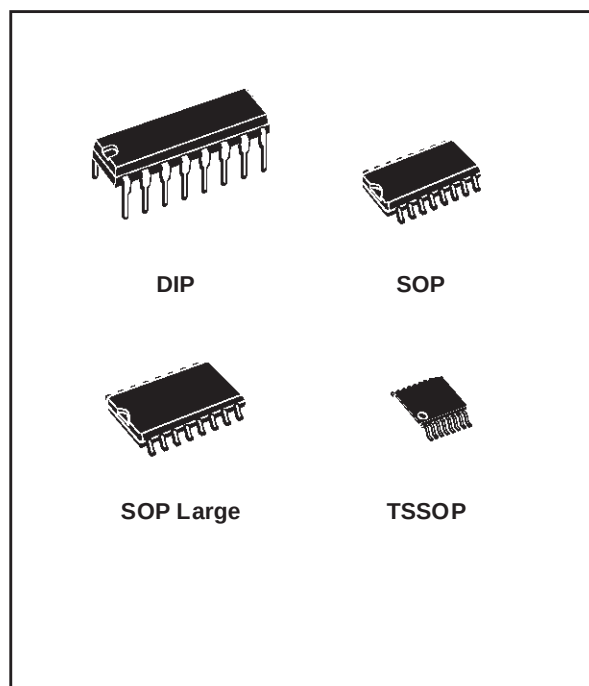
- 300 μ A SUPPLY CURRENT
- 300Kbps MINIMUM GUARENTEED DATA RATE
- 6V/ μ s MINIMUM GUARANTEED SLEW RATE
- MEET EIA/TIA-232 SPECIFICATIONS DOWN TO 3V
- AVAILABLE IN DIP-16, SO-16, SO-16 LARGE AND TSSOP16

DESCRIPTION

The ST3232 is a 3V powered EIA/TIA-232 and V.28/V.24 communication interface with low power requirements, high data-rate capabilities. ST3232 has a proprietary low dropout transmitter output stage providing true RS-232 performance from 3 to 5.5V supplies. The device requires only four small 0.1 μ F standard external capacitors for operations from 3V supply.

The ST3232 has two receivers and two drivers.

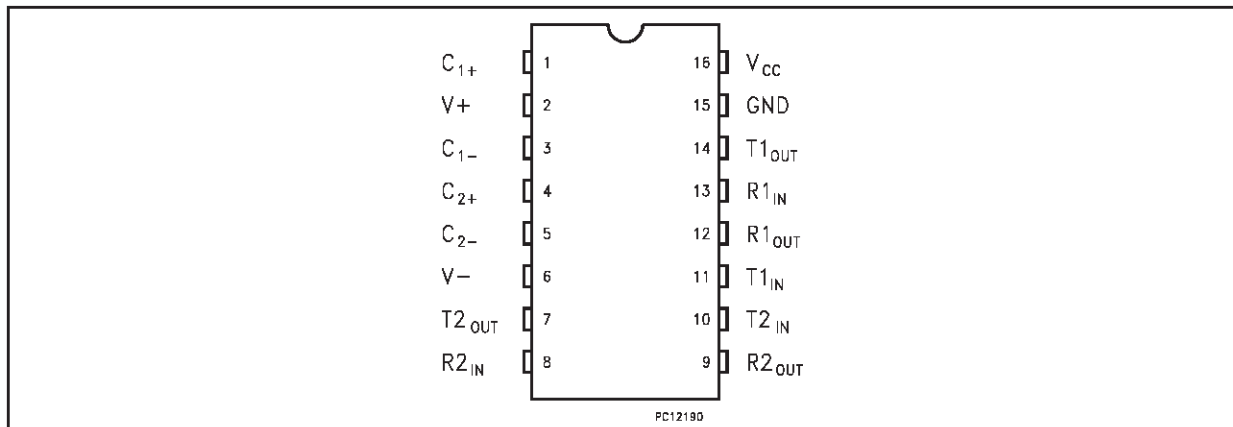
The device is guaranteed to run at data rates of 250Kbps while maintaining RS-232 output levels. Typical applications are Notebook, Subnotebook and Palmtop Computers, Battery Powered Equipment, Hand-Held Equipment, Peripherals and Printers.



ORDERING CODES

Type	Temperature Range	Package	Comments
ST3232CN	0 to 70 °C	DIP-16	25parts per tube / 40tube per box
ST3232BN	-40 to 85 °C	DIP-16	25parts per tube / 40tube per box
ST3232CD	0 to 70 °C	SO-16 (Tube)	50parts per tube / 20tube per box
ST3232BD	-40 to 85 °C	SO-16 (Tube)	50parts per tube / 20tube per box
ST3232CDR	0 to 70 °C	SO-16 (Tape & Reel)	2500 parts per reel
ST3232BDR	-40 to 85 °C	SO-16 (Tape & Reel)	2500 parts per reel
ST3232CW	0 to 70 °C	SO-16 Large (Tube)	49parts per tube / 25tube per box
ST3232BW	-40 to 85 °C	SO-16 Large (Tube)	49parts per tube / 25tube per box
ST3232CWR	0 to 70 °C	SO-16 Large (Tape & Reel)	1000 parts per reel
ST3232BWR	-40 to 85 °C	SO-16 Large (Tape & Reel)	1000 parts per reel
ST3232CTR	0 to 70 °C	TSSOP16 (Tape & Reel)	2500 parts per reel
ST3232BTR	-40 to 85 °C	TSSOP16 (Tape & Reel)	2500 parts per reel

PIN CONFIGURATION



PIN DESCRIPTION

PIN N°	SYMBOL	NAME AND FUNCTION
1	C ₁₊	Positive Terminal for the first Charge Pump Capacitor
2	V+	Doubled Voltage Terminal
3	C ₁₋	Negative Terminal for the first Charge Pump Capacitor
4	C ₂₊	Positive Terminal for the second Charge Pump Capacitor
5	C ₂₋	Negative Terminal for the second Charge Pump Capacitor
6	V-	Inverted Voltage Terminal
7	T _{2OUT}	Second Transmitter Output Voltage
8	R _{2IN}	Second Receiver Input Voltage
9	R _{2OUT}	Second Receiver Output Voltage
10	T _{2IN}	Second Transmitter Input Voltage
11	T _{1IN}	First Transmitter Input Voltage
12	R _{1OUT}	First Receiver Output Voltage
13	R _{1IN}	First Receiver Input Voltage
14	T _{1OUT}	First Transmitter Output Voltage
15	GND	Ground
16	V _{CC}	Supply Voltage

ABSOLUTE MAXIMUM RATINGS

Symbol	Parameter	Value	Unit
V _{CC}	Supply Voltage	-0.3 to 6	V
V+	Doubled Voltage Terminal	(V _{CC} - 0.3) to 7	V
V-	Inverted Voltage Terminal	0.3 to -7	V
V+ + V-		13	V
T _{IN}	Transmitter Input Voltage Range	-0.3 to 6	V
R _{IN}	Receiver Input Voltage Range	± 25	V
T _{OUT}	Transmitter Output Voltage Range	± 13.2	V
R _{OUT}	Receiver Output Voltage Range	-0.3 to (V _{CC} + 0.3)	V
t _{SHORT}	Transmitter Output Short to GND Time	Continuous	

Absolute Maximum Ratings are those values beyond which damage to the device may occur. Functional operation under these condition is not implied. V+ and V- can have a maximum magnitude of +7V, but their absolute addition can not exceed 13 V.

ELECTRICAL CHARACTERISTICS(C₁ - C₄ = 0.1μF, V_{CC} = 3V to 5.5V, T_A = -40 to 85°C, unless otherwise specified.Typical values are referred to T_A = 25°C)

Symbol	Parameter	Test Conditions	Min.	Typ.	Max.	Unit
I _{SUPPLY}	V _{CC} Power Supply Current	No Load V _{CC} = 3V or 5V T _A = 25°C		0.3	1	mA

LOGIC INPUT ELECTRICAL CHARACTERISTICS(C₁ - C₄ = 0.1μF, V_{CC} = 3V to 5.5V, T_A = -40 to 85°C, unless otherwise specified.Typical values are referred to T_A = 25°C)

Symbol	Parameter	Test Conditions	Min.	Typ.	Max.	Unit
V _{TIL}	Input Logic Threshold Low	T-IN (Note 1)			0.8	V
V _{TIH}	Input Logic Threshold High	V _{CC} = 3.3V V _{CC} = 5V	2 2.4			V V
I _{IL}	Input Leakage Current	T-IN		± 0.01	± 1	μA

Note 1: Transmitter input hysteresis is typically 250mV

TRANSMITTER ELECTRICAL CHARACTERISTICS(C₁ - C₄ = 0.1μF tested at 3.3V±10%, V_{CC} = 5V ± 10%, T_A = -40 to 85°C, unless otherwise specified.Typical values are referred to T_A = 25°C)

Symbol	Parameter	Test Conditions	Min.	Typ.	Max.	Unit
V _{TOUT}	Output Voltage Swing	All Transmitter outputs are loaded with 3KΩ to GND	± 5	± 5.4		V
R _{TOUT}	Transmitter Output Resistance	V _{CC} = V+ = V- = 0V V _{OUT} = ± 2V	300	10M		Ω
I _{TSC}	Output Short Circuit Current	V _{CC} = 3V to 5V V _{OUT} = ± 12V			± 60	mA

RECEIVER ELECTRICAL CHARACTERISTICS(C₁ - C₄ = 0.1μF tested at 3.3V±10%, V_{CC} = 5V ± 10%, T_A = -40 to 85°C, unless otherwise specified.Typical values are referred to T_A = 25°C)

Symbol	Parameter	Test Conditions	Min.	Typ.	Max.	Unit
V _{RIN}	Receiver Input Voltage Operating Range		-25		25	V
V _{RIL}	RS-232 Input Threshold Low	T _A = 25°C V _{CC} = 3.3V T _A = 25°C V _{CC} = 5V	0.6 0.8	1.2 1.5		V V
V _{RIH}	RS-232 Input Threshold High	T _A = 25°C V _{CC} = 3.3V T _A = 25°C V _{CC} = 5V		1.5 1.8	2.4 2.4	V V
V _{RIHYS}	Input Hysteresis			0.3		V
R _{RIN}	Input Resistance	T _A = 25°C	3	5	7	KΩ
V _{ROL}	TTL/CMOS Output Voltage Low	I _{OUT} = 1.6mA			0.4	V
V _{ROH}	TTL/CMOS Output Voltage High	I _{OUT} = -1mA	V _{CC} -0.6	V _{CC} -0.1		V

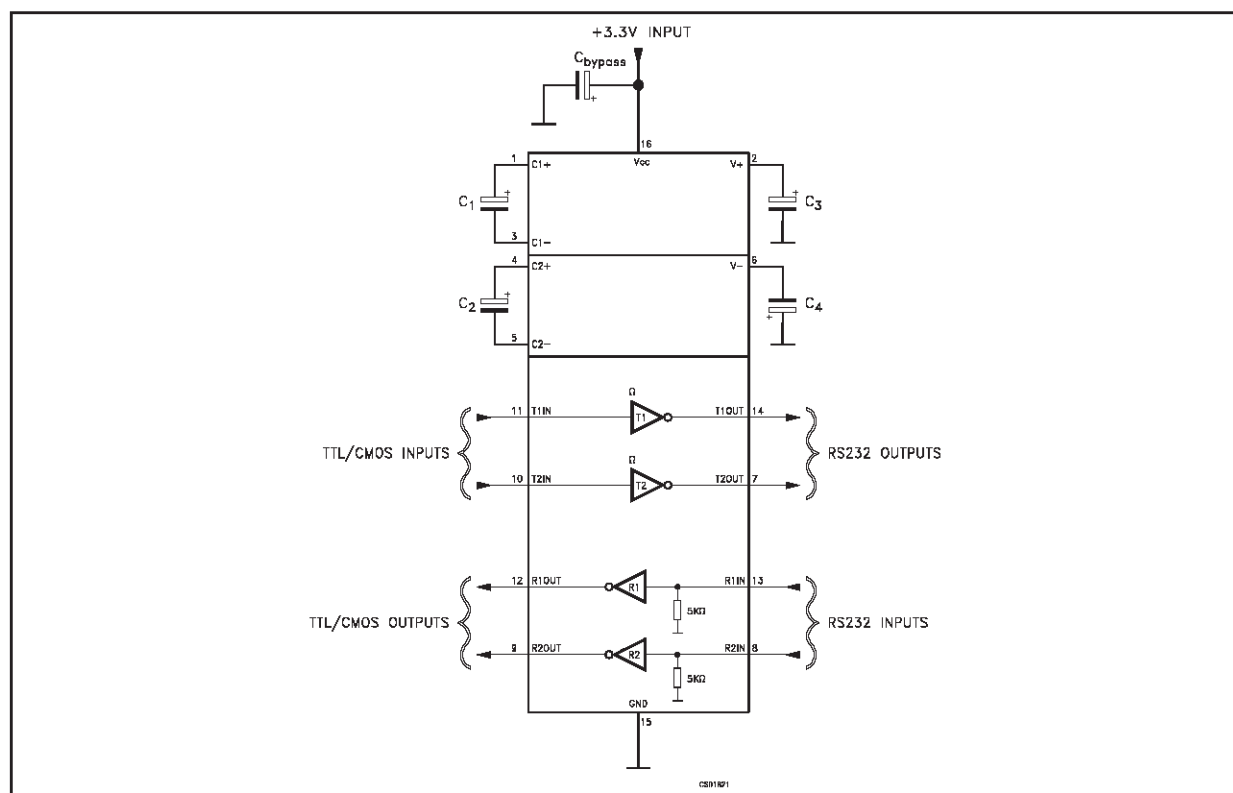
TIMING CHARACTERISTICS

($C_1 - C_4 = 0.1\mu\text{F}$, $V_{CC} = 3\text{V to } 5.5\text{V}$, $T_A = -40 \text{ to } 85^\circ\text{C}$, unless otherwise specified.)

Typical values are referred to $T_A = 25^\circ\text{C}$)

Symbol	Parameter	Test Conditions	Min.	Typ.	Max.	Unit
D_R	Data Transfer Rate	$R_L = 3\text{K}\Omega$ $C_{L2} = 1000\text{pF}$ one trasmitter switching	300	400		Kbps
t_{PHLR} t_{PLHR}	Propagation Delay Input to Output	$R_{XIN} = R_{XOUT}$ $C_L = 150\text{pF}$		0.2		μs
$ t_{PHLT} - t_{THL} $	Transmitter Propagation Delay Difference	(Note 1)		100		ns
$ t_{PHLR} - t_{THR} $	Receiver Propagation Delay Difference			50		ns
S_{RT}	Trnasition Slew Rate	$T_A = 25^\circ\text{C}$ $R_L = 3\text{K}\Omega \text{ to } 7\text{K}\Omega$ $V_{CC} = 3.3\text{V}$ measured from $+3\text{V}$ to -3V or -3V to $+3\text{V}$ $C_L = 150\text{pF}$ to 1000pF $C_L = 150\text{pF}$ to 2500pF	6 4		30 30	$\text{V}/\mu\text{s}$ $\text{V}/\mu\text{s}$

Transmitter Skew is measured at the transmitter zero cross points

APPLICATION CIRCUITS**CAPACITANCE VALUE (μF)**

C1	C2.	C3	C4	Cbypass
0.1	0.1	0.1	0.1	0.1

TYPICAL PERFORMANCE CHARACTERISTICS (unless otherwise specified $T_j = 25^\circ\text{C}$)

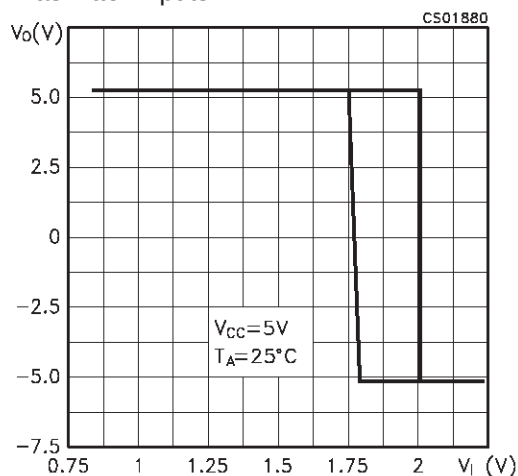
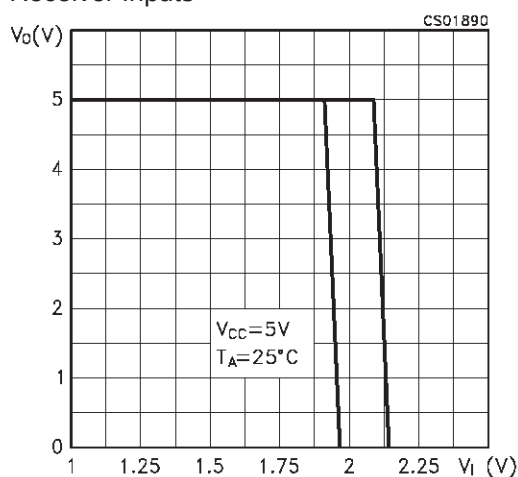
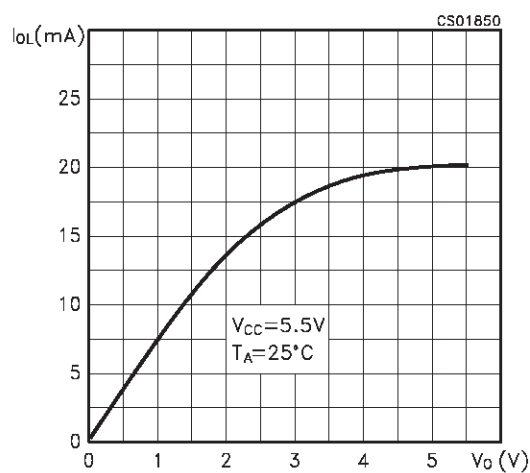
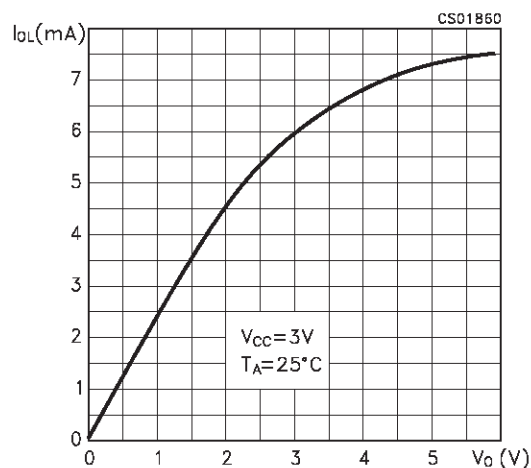
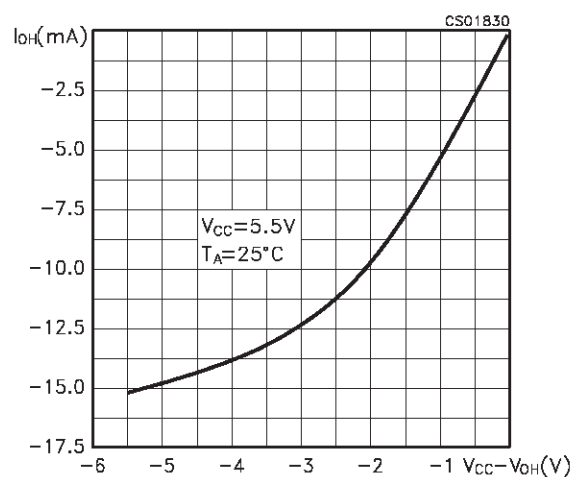
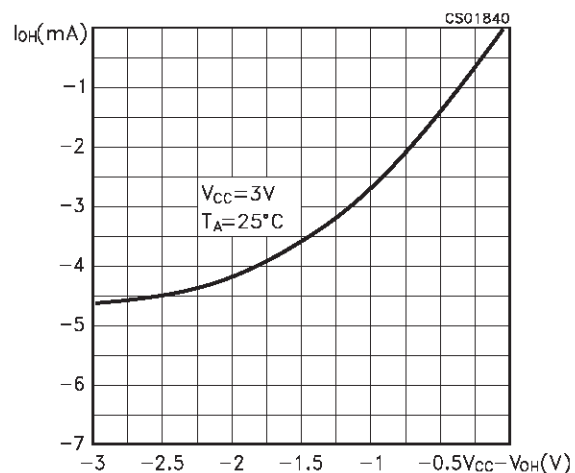
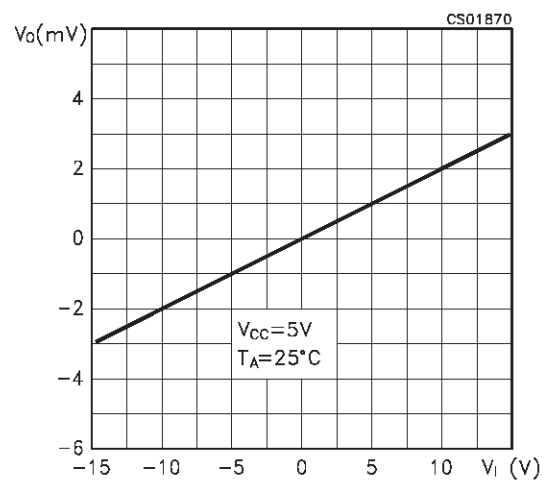
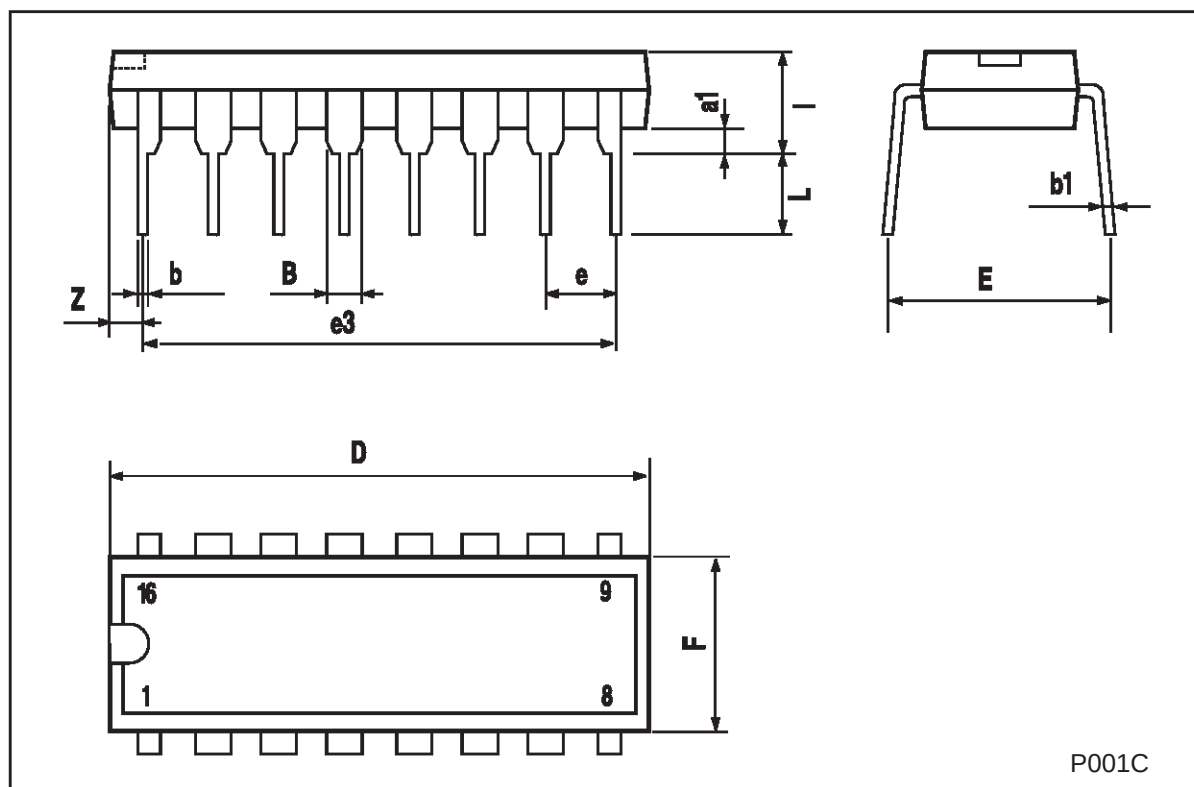
Figure 1 : Driver Voltage Transfer Characteristics for Transmitter Inputs

Figure 2 : Driver Voltage Transfer Characteristics for Receiver Inputs

Figure 3 : Output Current vs Output Low Voltage

Figure 4 : Output Current vs Output Low Voltage

Figure 5 : Output Current vs Output High Voltage

Figure 6 : Output Current vs Output High Voltage


Figure 7 : Receiver Input Resistance

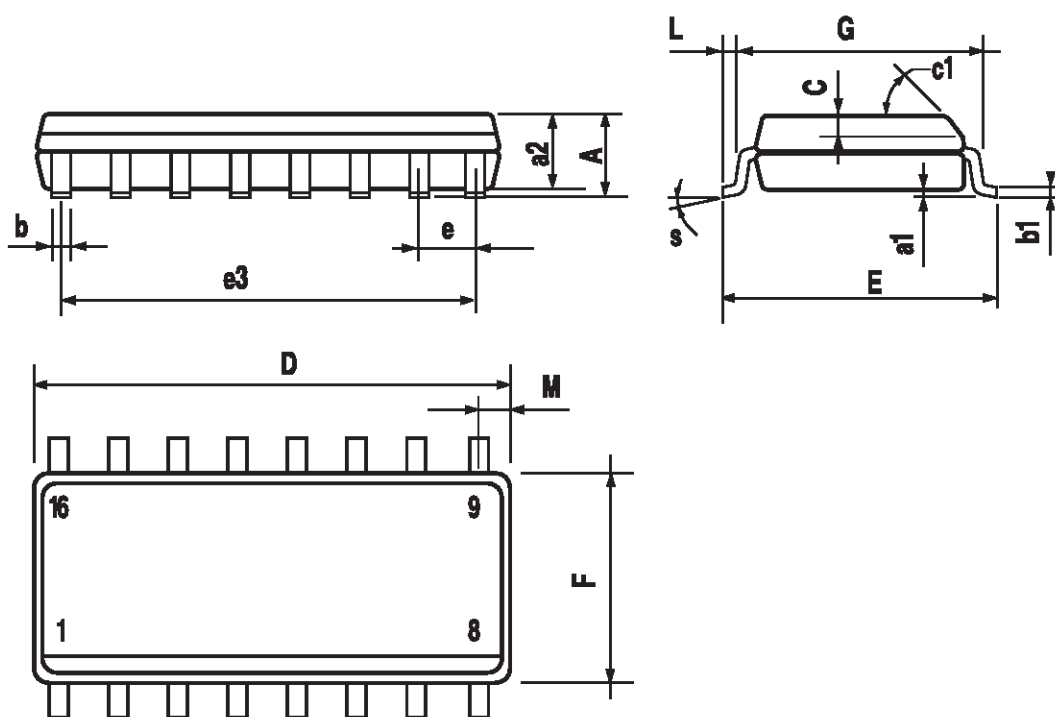
Plastic DIP-16 (0.25) MECHANICAL DATA

DIM.	mm			inch		
	MIN.	TYP.	MAX.	MIN.	TYP.	MAX.
a1	0.51			0.020		
B	0.77		1.65	0.030		0.065
b		0.5			0.020	
b1		0.25			0.010	
D			20			0.787
E		8.5			0.335	
e		2.54			0.100	
e3		17.78			0.700	
F			7.1			0.280
I			5.1			0.201
L		3.3			0.130	
Z			1.27			0.050



SO-16 MECHANICAL DATA

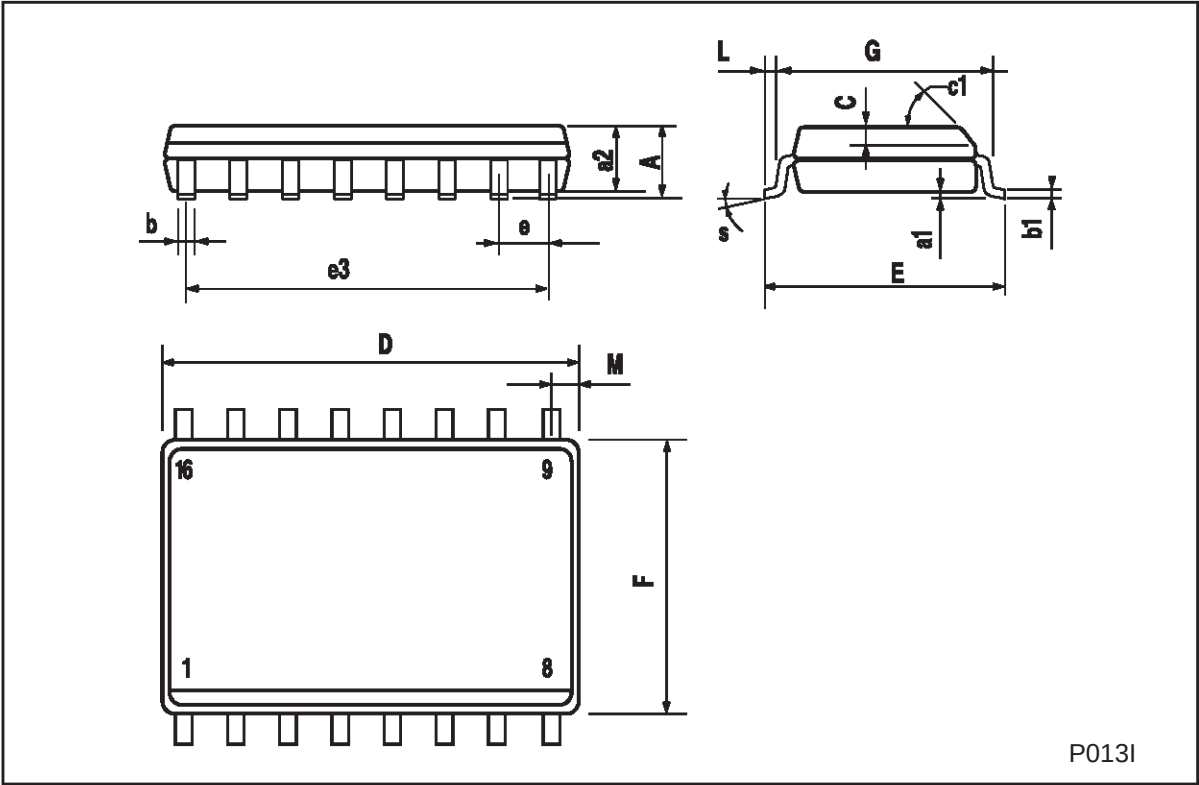
DIM.	mm			inch		
	MIN.	TYP.	MAX.	MIN.	TYP.	MAX.
A			1.75			0.068
a1	0.1		0.2	0.004		0.007
a2			1.65			0.064
b	0.35		0.46	0.013		0.018
b1	0.19		0.25	0.007		0.010
C		0.5			0.019	
c1	45 (typ.)					
D	9.8		10	0.385		0.393
E	5.8		6.2	0.228		0.244
e		1.27			0.050	
e3		8.89			0.350	
F	3.8		4.0	0.149		0.157
G	4.6		5.3	0.181		0.208
L	0.5		1.27	0.019		0.050
M			0.62			0.024
S	8 (max.)					



P013H

SO16L MECHANICAL DATA

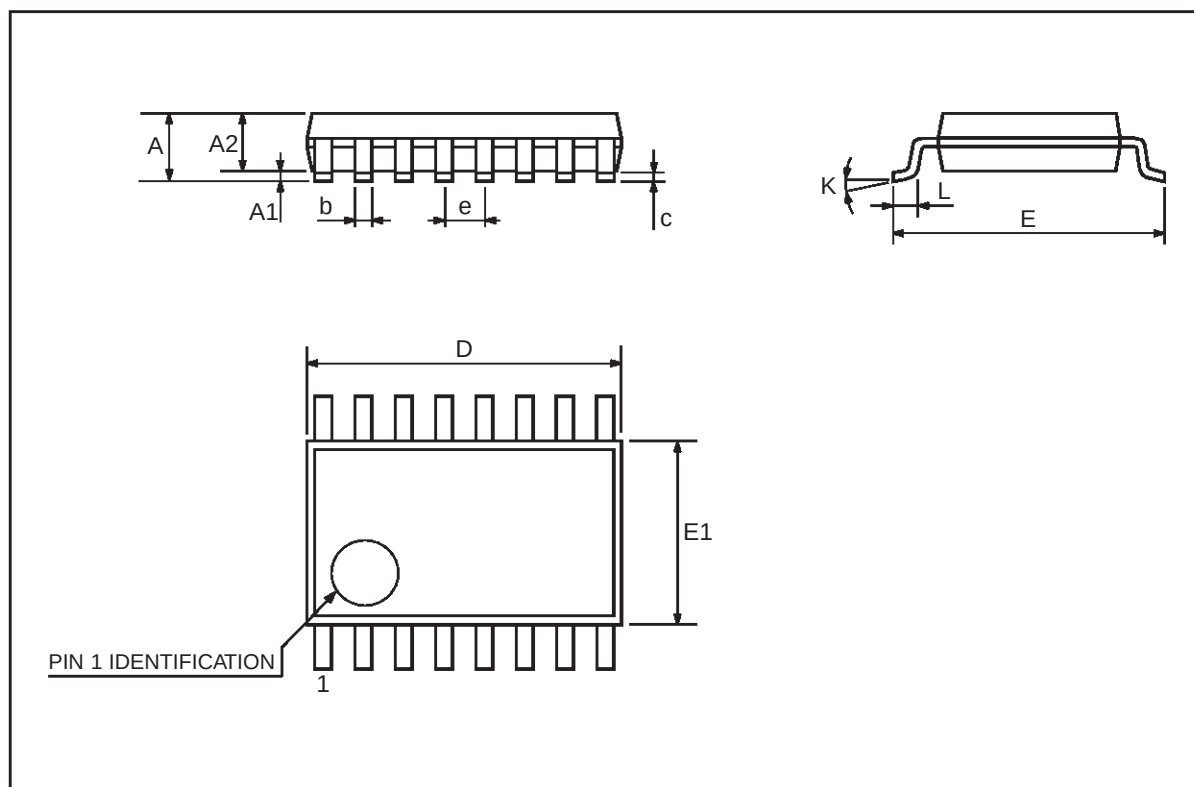
DIM.	mm			inch		
	MIN.	TYP.	MAX.	MIN.	TYP.	MAX.
A			2.65			0.104
a1	0.1		0.2	0.004		0.008
a2			2.45			0.096
b	0.35		0.49	0.014		0.019
b1	0.23		0.32	0.009		0.012
C		0.5			0.020	
c1	45 (typ.)					
D	10.1		10.5	0.397		0.413
E	10.0		10.65	0.3.93		0.419
e		1.27			0.050	
e3		8.89			0.350	
F	7.4		7.6	0.291		0.300
L	0.5		1.27	0.020		0.050
M			0.75			0.029
S	8 (max.)					



P013I

TSSOP16 MECHANICAL DATA

DIM.	mm			inch		
	MIN.	TYP.	MAX.	MIN.	TYP.	MAX.
A			1.1			0.433
A1	0.05	0.10	0.15	0.002	0.004	0.006
A2	0.85	0.9	0.95	0.335	0.354	0.374
b	0.19		0.30	0.0075		0.0118
c	0.09		0.20	0.0035		0.0079
D	4.9	5	5.1	0.193	0.197	0.201
E	6.25	6.4	6.5	0.246	0.252	0.256
E1	4.3	4.4	4.48	0.169	0.173	0.176
e		0.65 BSC			0.0256 BSC	
K	0°	4°	8°	0°	4°	8°
L	0.50	0.60	0.70	0.020	0.024	0.028



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